KHYATI KIYAWAT

khyati@virginia.edu | Khyati Kiyawat

RESEARCH FOCUS

Interests: Computer Architecture, Processing-in/near-Memory (PIM/PNM), LLM Inference

My research tackles the LLM Inference memory bottleneck by pioneering chiplet-based PIM and computational CXL devices. By co-designing hardware accelerators and performance modeling frameworks, my work enables cost- and energy-efficient solutions for next-generation AI systems.

EDUCATION

University of Virginia, VA, USA *2022 - 2028 (expected)*
Computer Science Ph.D. Student (*Advised by Prof. Kevin Skadron*) GPA: 3.88/4.0

Indian Institute of Technology Roorkee, India *2016 - 2020*
B.Tech, Department of Electronics and Communication GPA: 9.23/10.0

INDUSTRY EXPERIENCE

Memory Solution Research Intern | Samsung Semiconductors, USA *May 2026 - Aug 2026*

- Mentors: Gongjin Sun, Caroline Kahn
- Characterized the performance of Samsung's CXL-based memory devices as a KV cache offload tier for LLM inference using LMCache and compared them against System DDR and SSD baselines.

Processor Architecture Research Intern | Intel, India *May 2024 - Jan 2025*

- Mentors: Shankar Balachandran, Anant V. Nori, Sreenivas Subramoney
- Designed a framework for tensor tracking in GenAI workloads enabling rapid exploration of lossless KV cache compression in LLCs in Xeon Processors.

Front-end Design Verification Engineer | Texas Instruments, India *Summer 2019, July 2020 - July 2022*

- Verified SoC integration of registers, communication peripherals, and co-processors for C2000 real-time MCUs.
- Handled critical silicon debugs for functional test patterns on Automatic Test Equipment (ATE).
- Developed a Perl-based automation framework to auto-generate RTL for memory-mapped registers and the verification test cases thereby significantly reducing manual efforts and errors by 80%.

PUBLICATIONS

- **Reimagining LLM Inference Infrastructure with Memory-Centric KV Cache Servers**
  Khyati Kiyawat, Kevin Skadron. *HotInfra, 2026 (co-located with ISCA 2026).* *[link]*
- **HARMONI: Hierarchical ARchitecture MOdeling for LLMs with Near/In Memory Computing**
  Khyati Kiyawat, Yasas Seneviratne, Zhenxing Fan, Morteza Baradaran, Kevin Skadron. *ISPASS, 2026.* *[link]*
- **Sangam: A Chiplet-Based DRAM-PIM Accelerator with CXL Integration for LLM Inferencing**
  Khyati Kiyawat, Zhenxing Fan, Yasas Seneviratne, Morteza Baradaran, Akhil Shekar, Zihan Xia, Mingu Kang, Kevin Skadron. *arXiv, Nov 2025.* *[link]*
- **TriPIM — Exact Triangle Counting on UPMEM PIM for Graph Analytics.** *(LBR) MemSys, 2025.* *[link]*
  Morteza Baradaran, **Khyati Kiyawat**, Akhil Shekar, Abdullah Mughrabi, Kevin Skadron.
- **Membrane: Accelerating Database Analytics with DRAM-Based PIM Filtering and Schema Denormalization.** *TACO, 2025.* *[link]*
  Akhil Shekar, Kevin Gaffney, Martin Prammer, **Khyati Kiyawat**, Lingxi Wu, Helena Caminal, Zhenxing Fan, Yimin Gao, Ashish Venkat, José F. Martínez, Jignesh Patel, Kevin Skadron.

- **Architectural Modeling and Benchmarking for Digital DRAM PIM.** *IISWC, 2024. [link]*
  F. Siddique, D. Guo, Z. Fan, M. Gholamrezaei, M. Baradaran, A. Ahmed, H. Abbot, K. Durrer, K. Nandagopal, E. Ermovick, **K. Kiyawat**, B. Gul, A. Mughrabi, A. Venkat, K. Skadron.
- **An Efficient Scaling-Free Folded Hyperbolic CORDIC Design Using a Novel Low-Complexity Power-of-2 Taylor Series Approximation.** *TVLSI, 2023. [link]*
  Anu Verma, **Khyati Kiyawat**, Bishnu P. Das, Pramod K. Meher.
- **Real-Time Minimum Energy Point Tracking Using a Predetermined Optimal Voltage Setting Strategy.** **Khyati Kiyawat**, Yutaka Masuda, Jun Shiomi, Tohru Ishihara. *ISVLSI, 2020. [link]*

UNDERGRADUATE RESEARCH PROJECTS

Scale-free Hyperbolic CORDIC Architecture | IIT Roorkee, India *Aug. 2019 - Nov. 2021*
- *Advisor:* Prof. Bishnu P Das, ECE Department
- Designed a low-latency CORDIC architecture to compute $\sinh$ and $\cosh$ functions with desired precision.

Power optimization of RISC-V PULPino | Nagoya University, Japan *Dec. 2019 - Feb. 2020*
- *Advisor:* Prof. Tohru Ishihara, Graduate School of Informatics
- Proposed a real-time Minimum Energy Point tracking method by analyzing power consumption on RISC-V-based PULPino processor with at most 3.1% energy loss.

Image Processing based fish seed counting | IIT Bombay, India *May 2018 - July 2018*
- *Advisor:* Prof. Maryam S Baghini, VLSI Design Lab
- Proposed a novel data-acquisition apparatus to create a training dataset for a crowd-counting-based CNN model.

TEACHING AND MENTORING

Graduate Teaching Assistant | University of Virginia, USA
- Computer Systems and Organization I (CS 2130) - *Fall 2023*
- Computer Systems and Organization II (CS 3130) - *Spring 2024*
- CPU/GPU Memory Systems (CS 6501 Special Topics) - *Spring 2025, Spring 2026*

Graduate Mentorship Programs | University of Virginia, USA *2024 - Present*
- Mentoring an undergraduate student on KV cache optimization research for Agentic AI (with Dr. Skadron).
- Mentored a junior PhD student on CS doctoral program navigation (CS Graduate Student Group).
- Mentored a CS undergraduate on graduate school applications (Engineering Grad School Mentor Program).

Rise Together | Charlottesville, Virginia *Oct. 2023 - April 2024*
- Assisted high-school students in the US for undergraduate college applications and transition process.

Student Mentorship Programme | IIT Roorkee, India *Aug. 2018 - May 2020*
- Mentored 12 first-year students in their transition and adjustment to the institute's culture.

AWARDS AND ACHIEVEMENTS

- Recognized by *Semiconductor Research Corporation (SRC)* through *Student Spotlight in September 2026.*
- *Rewarded for the* Most Memorable Internship Project *at Intern Showcase, Samsung Semiconductors, 2026.*
- *Selected to participate in Computing Research Association-Widening Participation (CRA-WP) Grad Cohort Workshop for Women, San Francisco, USA, 2023.*
- *Awarded with* UVA Engineering Distinguished Fellowship 2022 *given to the top Ph.D. applicants.*
- *Semi-finalist in* Swadeshi Microprocessor Challenge 2020-21 *to build a RISC-V based prototype for autonomous emergency distress system in automobiles.*
- *Recipient of Honda Y-E-S Award 2018 awarded to students showing research potential in sustainable technology and also selected for Y-E-S Plus Scholarship 2019 to undergo an internship in Japan.*

- *Quarter-finalist in* India Innovation Challenge Design Contest (IICDC) 2018 *organized by Texas Instruments for developing a navigation aid system for fishermen near international waters.*
- *Awarded with 1988 Batch Award 2017 given to one undergraduate student based on the academic merit.*
- *Won First prize in* Spectrum *(circuit design) and Third prize in* Ideaz *(innovative applications of IoT) at Annual Tech Fest, IIT Roorkee, 2017.*

SERVICE

Conference Reviewer | ISCA *2024, 2025*
- Served as Artifact Evaluation Committee Member.

Steering Committee Member | Computer Architecture Student Association - CASA *2025*
- Leading the Computer Architecture Long-term Mentorship (CALM) program.

Systems Interest Group (SIG) | University of Virginia *Nov. 2022 - May 2024*
- Organized cross-departmental systems research forum, facilitating faculty-student discussions.

Unnat Bharat Abhiyan | IIT Roorkee *Sep. 2016 - May 2018*
- Contributed to the Govt. of India's initiative, leveraging the technical expertise of IITs to drive rural development.

SOPAN | IIT Roorkee, India *Aug. 2018 - May 2020*
- Conducted weekly educational outreach sessions for primary and middle school students in a nearby rural village.

PRESENTATIONS

- Reimagining LLM inference with Memory-centric Servers at *ORNL Core Universities AI Workshop 2026*.
- Open-Source Processing-In-Memory (PIM) Architecture Design through FPGA Emulation: A Case Study Modeling Sieve at *Open-Source Computer Architecture Research Workshop. [OSCAR @ ISCA 2023]*
- PIMperf: Analytical Modeling for SIMD Instructions on PIM at *CAPWIC 2024.*
- Chiplet-based DRAM PIM Architecture for LLM Inference at *TECHCON 2024.*